the qubit. It may be thought of as the quantum counterpart of the classical bit, the fundamental unit of information in a digital computer. However, the similarity between a bit and a qubit ends there.

A classical bit can store only one of two possible values: 0 or 1. A qubit, on the other hand, can exist in the state $|0\rangle$ (read as 'ket zero'), $|1\rangle$ (read as 'ket one'), or in a superposition of these two states. This idea also explains the title of this article, 'From Zero to Ket Zero'.

For visualisation and study purposes, the state of a single qubit can be represented using the Bloch sphere, which is named after the physicist Felix Bloch, another Nobel laureate. Figure 1 shows the Bloch sphere representation of the state $|0\rangle$. The figure was generated using the Bloch sphere simulator available at *https://bloch.kherb.io/*, which is distributed under the MIT License. This simulator is a useful tool for studying the effect of various quantum gates on a single qubit. It should be noted that the Bloch sphere representation works only for single-qubit systems. When more than one qubit is involved, the state of the system exists in a much higher-dimensional space, making a simple geometric representation like the Bloch sphere impossible.

Now let us try to understand the value stored in a qubit. Before proceeding further, it is worth recalling

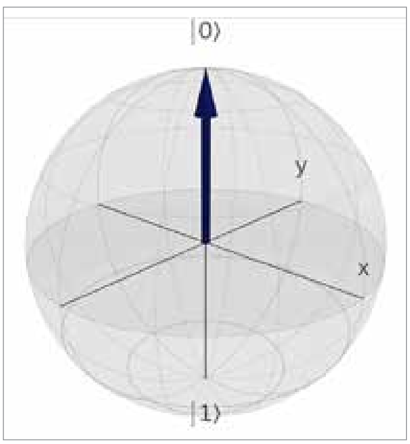

Figure 1: Bloch sphere representation of $|0\rangle$

a famous remark by Richard Feynman, which highlights the conceptual difficulty of quantum mechanics. Feynman once said, "I think I can safely say that nobody understands quantum mechanics." This statement reflects the inherent complexity of the subject and explains why, even today, relatively few people work in fields such as quantum computing.

Returning to our technical discussion, the state of a single qubit can be visualised using the Bloch sphere. In this representation, the state $|0\rangle$ corresponds to the north pole of the Bloch sphere (see Figure 1), while the state $|1\rangle$ corresponds to the south pole. However, the most important aspect of a qubit is that it can also exist in a superposition of these two states. In general, the state of a qubit can be written as $|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ where α and β are complex numbers known as probability amplitudes. These amplitudes satisfy the normalization condition $|\alpha\rangle^2+|\beta\rangle^2=1$. Here, $|\alpha\rangle^2$ represents the probability of measuring the state $|0\rangle$, while $|\beta\rangle^2$ represents the probability of measuring the state $|1\rangle$. For example, consider the qubit state $|\psi\rangle=\sqrt{(1/2)}|0\rangle+\sqrt{(1/2)}|1\rangle = \sqrt{(1/2)}(|0\rangle+|1\rangle)$. In this case, the probabilities of measuring $|0\rangle$ and $|1\rangle$ are both 1/2 because $|\alpha|^2=|\beta|^2=(\sqrt{(1/2)})^2=1/2$. We will explore these concepts in greater detail in later articles.

Now, let us take a moment to understand complex numbers, which play a central role in quantum computing. A complex number is a number of the form $z=a+bi$, where a and b are real numbers, and i is the imaginary unit defined by $i^2=-1$. Complex numbers allow us to represent quantities that have both magnitude and phase, which are essential in describing quantum states and their evolution.

Now a word about the notation $|0\rangle$, $|1\rangle$, and $|\psi\rangle$. This notation is known as bra–ket notation, or Dirac notation, and was introduced by the physicist and Nobel laureate Paul Dirac. As you may have already noticed, the story of quantum computing involves many Nobel Prize–winning physicists.

Two mathematical fields are particularly important for understanding quantum computing: linear algebra and probability theory. Throughout this series, we will cover the necessary mathematical concepts whenever possible. In cases where a detailed explanation is not feasible, the required topics will be clearly indicated so that you can study them separately.

For example, we have already briefly discussed complex numbers, which play a central role in quantum computing. As we move forward, it will also be essential to understand concepts related to matrices and matrix operations. In particular, familiarity with matrix multiplication will be necessary to follow some of the upcoming discussions. If you are not comfortable with these topics, it may be helpful to review them before proceeding further. The Wikipedia article titled 'Matrix (mathematics)' is a good starting point for learning about matrices and their properties.

In quantum computing, the state of a qubit is represented using complex probability amplitudes. This means that the coefficients appearing in the expression for a qubit state are complex numbers. For example, the two basic states of a qubit, $|0\rangle$ and $|1\rangle$, can be represented using vectors whose entries are complex numbers. In vector form, these basis states are written as $|0\rangle=[1 \quad 0]^T$ and $|1\rangle=[0 \quad 1]^T$. These two vectors form the computational basis for a single qubit. Any general qubit state can be expressed as a linear combination of these two basis states with complex coefficients.

Recall from twelfth-standard mathematics that a row vector is a matrix with a single row, while a column vector is a matrix with a single column. The transpose of a row vector A, denoted by A^T, is a column vector , and the transpose of a column vector is a row vector.